



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.Sc. EXAMINATION

COLLEGE: SCIENCE & TECHNOLOGY

DEPARTMENT: COMPUTER & INFORMATION SCIENCES

SESSION: 2020/2021

COURSE CODE: CSC311

COURSE TITLE: DISCRETE STRUCTURES

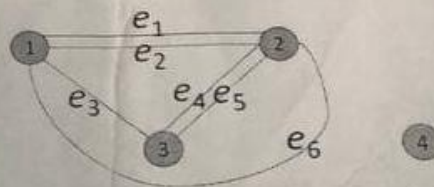
INSTRUCTION: ATTEMPT ANY THREE (3) QUESTIONS

SEMESTER: ALPHA

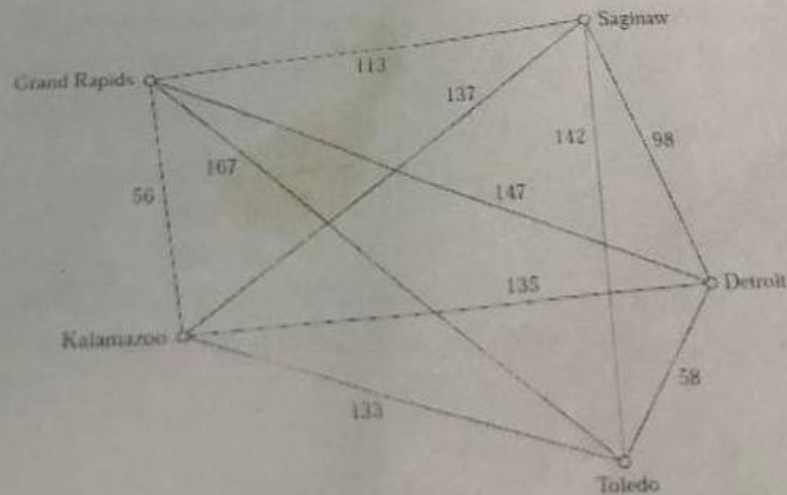
CREDIT UNIT: TWO (2)

TIME: 2 HOURS

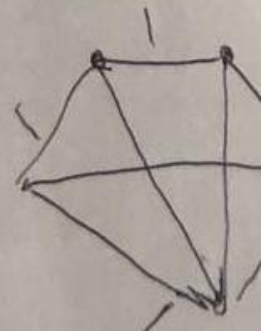
1. (a) Explain, with the aid of relevant examples, the types of undirected graph. (12marks) ✓
(b) Assuming you are required to set up a simple communication link between CSIS, CST Lab, CLR, CUCRID and Senate Building, how many of such simple communication links are possible? Support your answer with a diagrammatical illustration. (6½marks) ✓
(c) When is a Vertex said to be *Incident* to an Edge? Identify all the edges each of the vertices is incident to in the diagram below. (5marks) ✓



2. (a) A Christian missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance? (10marks)



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(b) Highlight, with two relevant examples each, four real-life scenarios where graph models are applicable. (6marks)

(c) Highlight the various key terminologies of a Rooted Tree. Support your answer with a diagrammatical illustration. (7½marks)

3. (a) Explain, with examples, what a Balanced M-ary Tree means? (7½marks)

(b) Suppose that someone starts a chain letter. Each person who receives the letter is asked to send it on to four other people. Some people do this, but others do not send any letters. How many people have seen the letter, including the first person, if no one receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not send it out? How many people sent out the letter? (8marks)

(c) Highlight, with a relevant example each, four application areas in computing where graph models are applicable. (8marks)

4. (a) Playoff between two teams consists of at most five games. The first team that wins three games wins the playoff. (4marks)

(i) Construct a tree diagram for five games. (3½marks)

(ii) In how many different ways can the playoff occur?

(b) With the aid of a diagram, explain the following types of function correspondences types:

(i) One-to-one, not onto (ii) Onto, not one-to-one (iii) One-to-one and onto (10 marks)

(iv) Neither one-to-one nor onto (v) Not a function

(c) Show that the following are logically equivalent (3marks)

(i) $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$. (3marks)

(ii) $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$

5. (a) Define and draw the truth table for each of the following logic proposition terminologies:

(i) Negation (ii) Conjunction (iii) Exclusive or (iv) Implication (6marks)

(v) Tautology (vi) Contingency

(b) Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? (5 marks)

(c) Consider these relations on the set of integers:

$R1 = \{(a, b) \mid a \leq b\}$

$R2 = \{(a, b) \mid a > b\}$

$R3 = \{(a, b) \mid a = b \text{ or } a = -b\}$

$R4 = \{(a, b) \mid a = b\}$

$R5 = \{(a, b) \mid a = b + 1\}$

$R6 = \{(a, b) \mid a + b \leq 3\}$ (5marks)

(i) List 5 ordered pairs in each of the relation.

(ii) Which of the relations are: (a) Reflexive (b) Symmetric (c) Transitive (6½marks)

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